

Aparna Dharmalingam

Functional Safety | ISO 26262 | ADAS & System Safety

M.Sc. candidate in Functional Safety Engineering with hands-on experience in applied research and automotive projects. Focus areas include HARA, ISO 26262, SOTIF, System Safety, and the safe development of automated driving functions. Currently seeking a full-time role in Functional Safety, System Safety, or ADAS Safety.



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📅 03.04.1998

EDUCATION

M.Sc. Functional Safety Engineering

University of Kassel

10/2020 – Present

Kassel, Germany

- Specialization: Automotive Functional Safety
- System Safety and system-level safety concepts

Bachelor of Engineering – Mechatronics

SCSVMV University

08/2015 – 04/2019

Kanchipuram, India

- Focus areas: MEMS, power electronics, control systems, and robotics

PROFESSIONAL EXPERIENCE

Student Research Assistant – Safety Engineering

Fraunhofer IESE

01/2026 – 04/2026

Kaiserslautern

- Transformed free-text hazardous events from HARA artifacts into standardized YAML safety models, creating a consistent, machine-readable basis for further safety analysis.
- Hazardous events to ODD parameters and operating conditions, improving the traceability of ASIL classification.
- Developed a Python-based tool chain to automate the processing of TOD elements and hazardous-event mappings, reducing manual data-preparation effort.

Working Student – Functional Safety (ADAS & AD)

Mingabyte GmbH

03/2024 – 09/2024

Munich

- Developed technical training material on automated-driving functions and ADAS Safety, enabling the structured communication of complex safety topics.
- Structured vehicle-level safety requirements and safety concepts, improving their clarity, consistency, and technical readability.
- Contributed as a co-assessor to a Functional Safety Assessment of a hydrogen fuel-cell system, supporting the systematic evaluation of safety-related evidence.

Working Student – Functional Safety

DEKRA Digital Products & Services

03/2023 – 09/2023

Stuttgart

- Tested an internal Functional Safety Assessment tool against ISO 26262 and IEC 61508, identifying gaps between tool logic and applicable requirements.
- Developed standards-based checklists and review workflows, supporting more consistent and repeatable internal safety assessments.
- Documented test findings and improvement proposals, contributing to further tool development and improved usability.

CORE COMPETENCIES

Safety Standards & Methods

ISO 26262 · IEC 61508 · IEC 61131-3 · IEC 61511 · ISO/SAE 21434 · ISO 21448 (SOTIF) · HARA · FMEA · FMEDA · HAZOP · FTA · Requirements Engineering

MBSE

System-level documentation, simulation, and design

Software & Tools

Ansys medini analyze · Python · NI LabVIEW · NI Multisim · PSIM · MATLAB/Simulink · ROS/SLAM · MS Office · PTC Creo · VBA · Git · YAML · B&R Automation Studio

Personal Strengths

Analytical thinking · structured approach · initiative · teamwork

ACADEMIC THESES

Master's Thesis – Developing a Criticality Taxonomy for Situation Analysis in Safety Engineering (in preparation)

Literature-based identification and structuring of criticality factors for driving situations, with planned application in SentryCraft for systematic scenario analysis.

Bachelor's Thesis – Sensor Assisted Braking System (09/2018 – 04/2019)

Modelled, developed, and validated a semi-electronic braking system on the Smart Trike hybrid vehicle; published under DOI: [10.1166/asem.2020.2606](https://doi.org/10.1166/asem.2020.2606)

ADDITIONAL EXPERIENCE

Autonomous Systems Engineer

Herkules Racing Team

02/2023 – 12/2024

Kassel

- Developed and tested ROS- and Python-based software modules for perception and planning on an autonomous competition vehicle.
- Supported sensor integration and safety-related troubleshooting at vehicle level.

Secretary – Technical University Club

Electro Infinity

09/2017 – 04/2019

Kanchipuram

- Organized a PCB design and manufacturing workshop and coordinated weekly technical meetings, including a classroom automation project.

LANGUAGES

English
C1

German
B1

Tamil
C2

INTERESTS

Motorcycling & Motorsport · Fencing · Singing · Cooking · Travel